Analytic Number Theory

I Dirichlet primo number theorem: If $\gcd(a,q)=1$, then $\sum_{p \equiv a \pmod q} \frac{1}{p}$ diverges

1. Dirichlet characters

Def I 1.1. A Dirichlet char mod q is a nonzero multiplicative function χ with period q that vanishes on n which are not relatively prime to q

We can prove that if $q = 2^{\alpha} p_1^{e_1} \cdots p_s^{e_s}$, then $\chi(n) = \chi(n, 2^{\alpha})\, \chi(n, p_1^{e_1}) \cdots \chi(n, p_s^{e_s})$

Tho construction of $\chi(n, p^{\alpha})$ (skipped) (See the notes in the last course of number theory